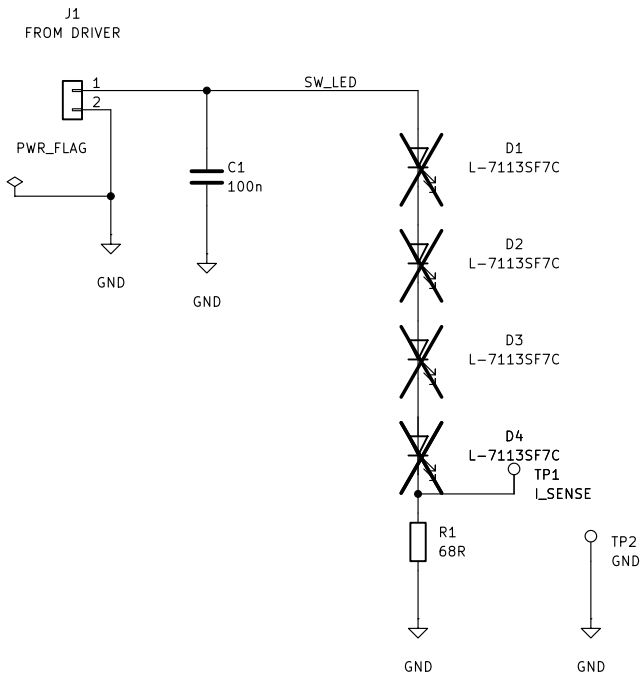




MARKER BOARD rev B – build 4 (A, B, C, D), all identical

Eight parts. This is the board that was already characterised, plus a connector – the constant-current sink of rev A is deleted. It bought about 5% marker-to-marker matching, which is smaller than the 3% run-to-run repeatability measured on the bench, and it cost 1.6 V of headroom, which was the entire reason rev A needed 12 V.

EMITTER Kingbright L-7113SF7C, 5 mm THT, GaAlAs, 850 nm (DSAB9772 V.16B).
Vf 1.4 typ / 1.6 max At If = 20 mA ONLY. Vf rises with current; -1.5 V each at 36 mA, so the string is ~5.9 V not 6.4 V.
If (DC) 50 mA absolute max
IfS (peak) 1 A, but ONLY at 1/100 duty and 10 us pulse width
Pd 80 mW. At 36 mA / 0.36% duty each LED averages 0.19 mW.
2 theta 1/2 20 deg – narrow. This is why the marker rings, and why tangential aiming matters more than shell diameter.

R1 SETS THE CURRENT. Same value on all four boards. Computed at 9 V less the driver's Schottky and bulk-cap droop, with the real Vf curve:
33R -> 61 mA 68R -> 36 mA (fitted, matches the bench build)
47R -> 48 mA 100R -> 26 mA
Measure across TP1-TP2 with a scope during a pulse: I = V(TP1) / R1.

SUPPLY SENSITIVITY is the price of dropping the CC sink: 0.3 V of battery sag moves the current 9%. Peak pixel moves with it, so re-target the exposure occasionally. It is a slow drift, not noise, and the bench data shows sigma changing only ~3% over a 20% exposure range – so it costs well under 1% of centroid accuracy. That is the trade, stated plainly.

CHECK THE WORKBOOK NUMBER. LED_characterisation.xlsx lists 38.2 mA for this build, computed from an assumed Vf of 1.6 V – which is the datasheet MAX at 20 mA, not the value at 38 mA. With the real curve, 9 V through 68R gives about 43 mA. Worth measuring once and correcting in the sheet.

D1..D4 are DNP so an assembler skips them – they are hand-soldered so the standoff to the diffuser can be set by hand. The pads are still required. Space them as widely as the shell allows and aim them TANGENTIALLY.

UCT MSc – Shannon Pitman		
Sheet: /		
File: WandMarker.kicad_sch		
Title: Wand marker – 4x Kingbright L-7113SF7C + one ballast resistor		
Size: A3	Date:	Rev: B
KiCad E.D.A. 10.0.4		Id: 1/1